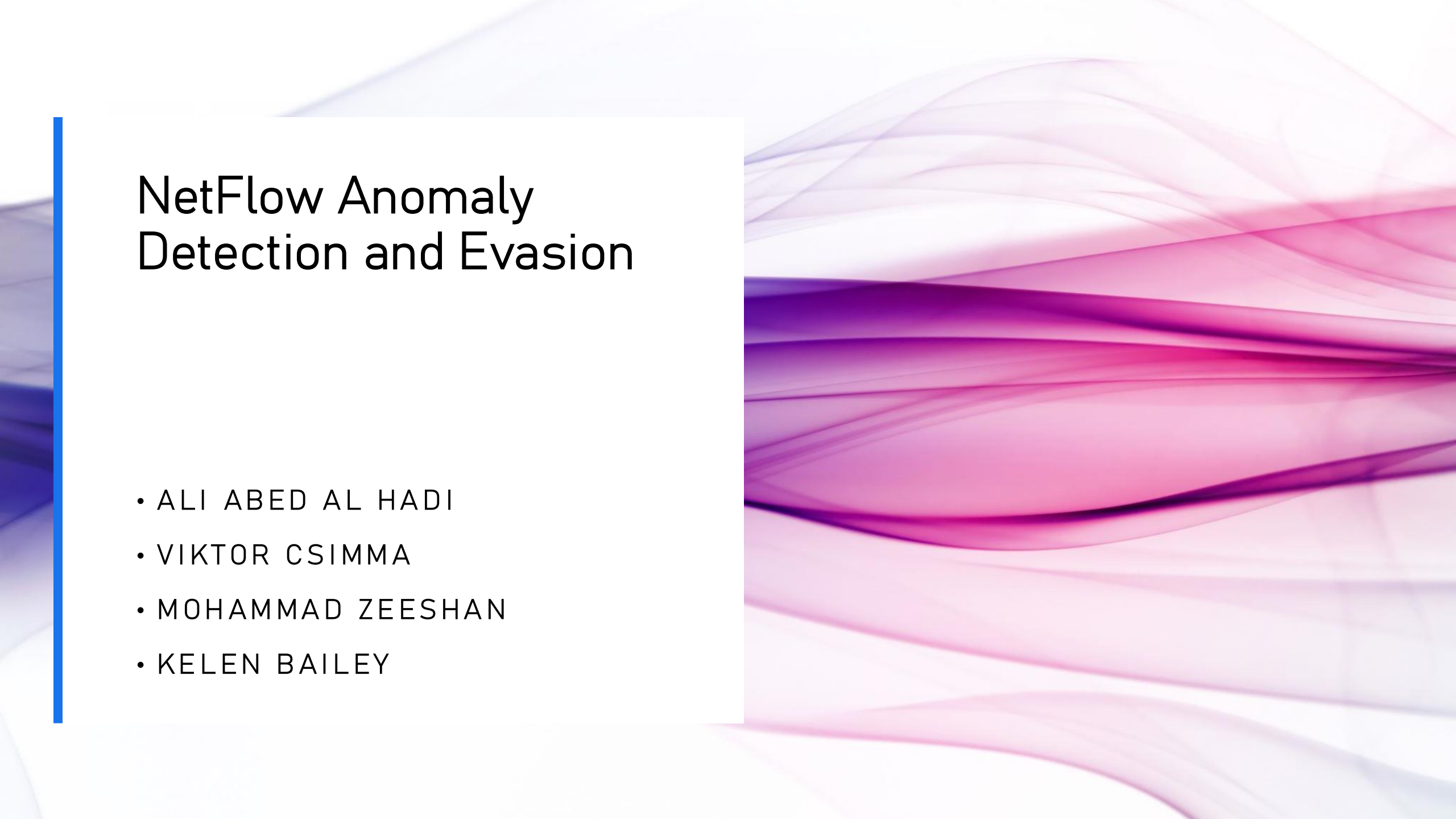




NetFlow Anomaly Detection and Evasion

- ALI ABED AL HADI
- VIKTOR CSIMMA
- MOHAMMAD ZEESHAN
- KELEN BAILEY

Scenario

	sport	dport	proto	state	dur	sbytes	spkts	label
1	10880	5060	udp	INT	0.000019	1388	2	0.881957031
2	42254	445	tcp	FIN	1.512013	10976	8	0.254432220
3	11233	25	tcp	FIN	1.547788	65540	66	0.873267082
4	42254	445	tcp	FIN	1.512013	10976	8	0.254432220
5	49346	53	tcp	CON	0.001111	132	2	0.000104099
6	23544	25	tcp	FIN	0.851111	115727	78	0.852064601
7	29230	445	tcp	FIN	0.852222	4700	4	0.395465102



Model types

- random forests
 - Ranger library
 - custom model
- other algorithms
 - SGDC, SVM, KNN, GaussianNB, DecisionTree, AdaBoost

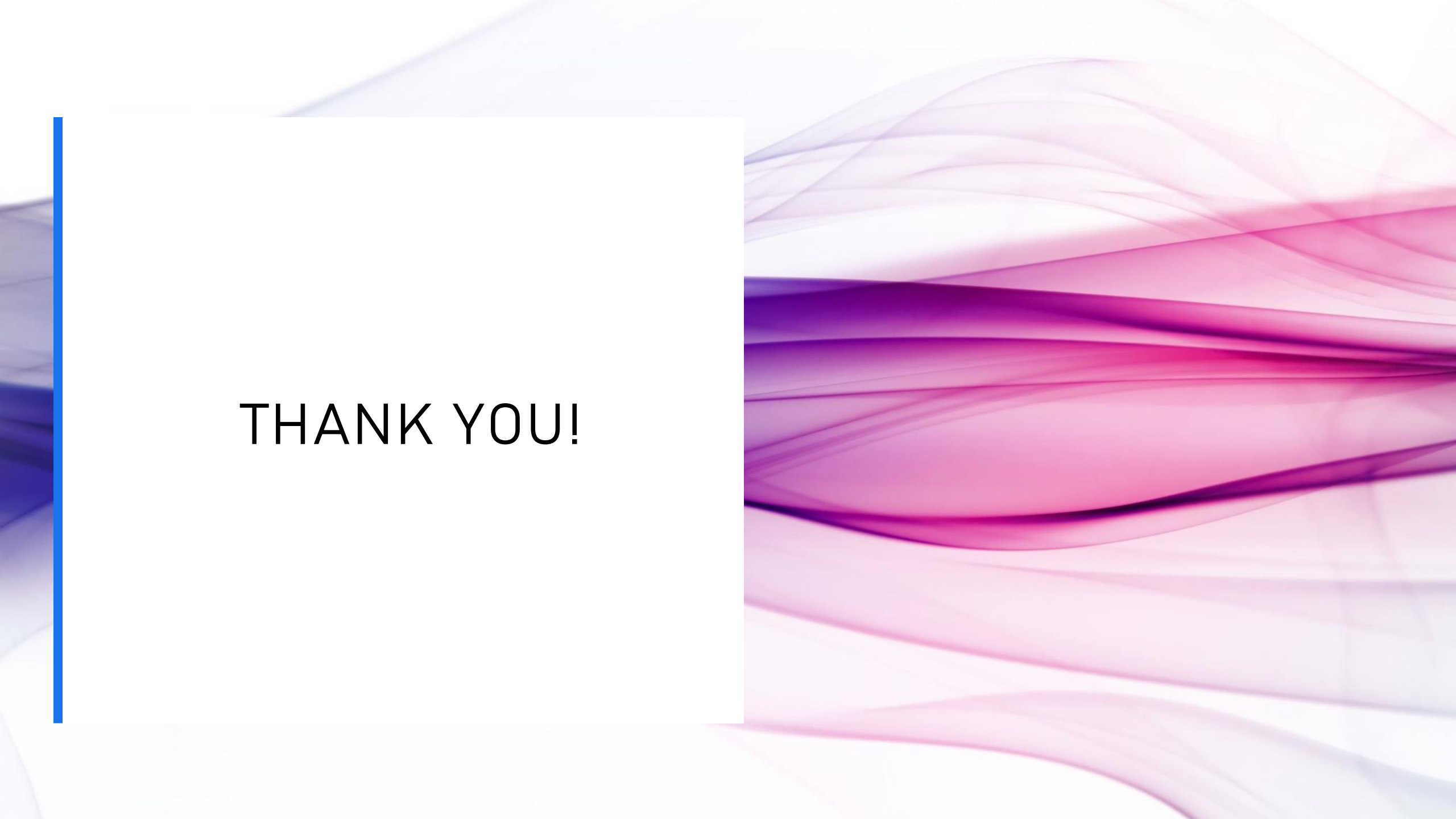
Attack types

- Whitebox:
 - training data poisoning
 - model poisoning
- Greybox:
 - Crafting Evasion records according to training data statistics
- Blackbox:
 - Stealth port and version scan



Who did what?

- Ali:
 - leadership and management
 - Greybox and blackbox evasions
- Viktor:
 - an R source code collection based on Ranger
 - model poisoning and backdoor attacks
- Mohammad:
 - a custom RF model with SMOTE
 - an interactive web application
- Kelen:
 - comparison of resilience with that of other, non-RF algorithms



THANK YOU!